

# Survival in the Era of High Uncertainty: The Cognitive Shift from Line-by-Line Coding to Managing Certainty

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As AI takes over more tasks, where does human value lie? Let's look at the industry frontier: in 2026, multi-agent systems have begun to land. OpenAI iterated the new **GPT-5.3 Codex** using agents, and Claude's **Opus 4.6** significantly enhanced its agentic capabilities, becoming the premier base model for **Claude Cowork**—a product launched this January for non-programmers. As AI systems grow more complex, they handle not just coding, but the entire software engineering (SE) lifecycle: deployment, testing, O&M, and monitoring. This expansion is rapidly touching every sector—entertainment, education, office work, and daily life.

As systems become more complex and capable, uncertainty increases exponentially. In the real business world, iteration speed is life. With AI doing 80% of the work, the demand for high quality and rapid iteration has skyrocketed. However, humans struggle to adapt to this "High Uncertainty" because the volume of AI-generated content in short timeframes has exceeded the physiological limits of human oversight.

The core value of humans in the future lies in **the ability to manage certainty within high uncertainty**. Traditional SE was always about this, but AI has intensified the challenge. In large-scale systems, we must trust robust testing rather than line-by-line analysis. No human can analyze every line anymore. This is counter-intuitive to beginners who think they must "understand every line," but in real-world scenarios, that is simply impossible. AI has made this reality unavoidable. Therefore, the best way to manage high uncertainty is the same as in traditional SE: **excellent testing**. In the AI era, you don't even need to write the tests yourself; you need the experience to direct AI to build them. Human value now lies in the ability to establish **reusable verification systems** and managing uncertainty in extremely short windows.

The disruption of SE by AI is a fait accompli. AI is even impacting the open-source ecosystem. Some argue AI might "ruin" open source by reducing meaningful human interaction. While **"Vibe Coding"** boosts short-term productivity, it might degrade the overall system quality in the long run. On GitHub, humans are already losing ground to AI. According to SemiAnalysis, Anthropic's **Claude Code** already accounts for 4% of public commits and is projected to reach 20% of daily commits by the end of 2026.

People now have a much higher threshold for what constitutes a "good product." In the future, the competition won't be about who builds a "toy" faster, but who can achieve extreme product refinement in the shortest time. First-mover advantage is yielding to **product experience**. This aligns with our economic shift: the past 30 years were about making money from development; the next 30 will be about winning through "struggle" and superior experience. Therefore, a core human value will be the **practice of perfection and high standards**.

Many can build an app in a few days now, but the subsequent refinement—security (rate limiting, IP blacklists, bot protection, CSRF), scalability, and maintenance—is where the real pain (and value) lies. Even if AI makes documents and tests easy to generate, the **connection and correction** of these elements still demand massive mental effort. While "premature optimization is the root of all evil" for an MVP, the future belongs only to those who can deliver high-quality, polished products.

These rules and insights may change in a few years. It's better not to overthink the next five years, but rather to plan and adapt within a five-year horizon. Constant learning and thinking are the only ways to handle this rapid transformation. This, perhaps, is the ultimate core value of a human.

# 我们如何利用 Codex 训练并部署 GPT-5.3-Codex

Codex 近期实现的快速进步，建立在 OpenAI 各个团队跨越数月乃至数年的研究项目成果之上。这些研究项目正由 Codex 加速推动；OpenAI 的许多研究员和工程师表示，他们如今的工作性质与短短两个月前相比已发生了根本性的变化。即使是 GPT-5.3-Codex 的早期版本，也展现出了卓越的能力，使我们的团队能够利用这些早期版本来改进训练，并支持后续版本的部署工作。

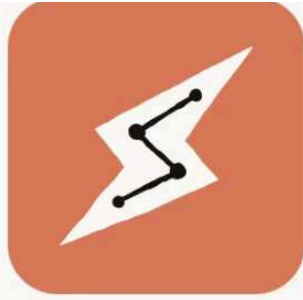
Codex 可用于极其广泛的任務，以至于很难完全列举它帮助我们团队的所有方式。举例来说，研究团队利用 Codex 监控并调试了本次发布的训练运行。它对研究的加速不仅限于调试基础设施问题：它还帮助追踪了整个训练过程中的模式，对交互质量进行了深度分析，提出了修复方案，并为人类研究员构建了功能丰富的应用程序，以便精确了解该模型的行为与以往模型相比有何差异。

工程团队利用 Codex 优化并调整了 GPT-5.3-Codex 的测试框架 (harness)。当我们开始发现影响用户的奇特边缘案例时，团队成员利用 Codex 识别出了上下文渲染中的漏洞，并找到了导致低缓存命中率的根本原因。在整个发布过程中，GPT-5.3-Codex 持续帮助团队动态缩放 GPU 集群，以应对流量激增并保持延迟稳定。

在 Alpha 测试期间，一位研究员想要了解 GPT-5.3-Codex 在每轮对话中多完成了多少工作，以及相关的生产力差异。GPT-5.3-Codex 构思了几个简单的正则表达式分类器，用以评估澄清请求、用户正向与负面反馈以及任务进展的频率；随后，它将这些分类器在大规模会话日志上运行，并生成了一份带有结论的报告。使用 Codex 进行构建的人员感到更加满意，因为智能体能更好地理解他们的意向，且每轮对话的进展更多，所需的澄清问题更少。

由于 GPT-5.3-Codex 与其前代产品差异巨大，Alpha 测试的数据呈现出许多异常且违背直觉的结果。团队中的一位数据科学家与 GPT-5.3-Codex 合作构建了新的数据流水线，并以远超常规看板工具的方式，更丰富地实现结果可视化。随后，他们与 Codex 共同分析了结果，Codex 在不到三分钟的时间内，简明扼要地总结了数千个数据点中的关键洞察。

单独来看，所有这些任务都是 Codex 如何帮助研究人员和产品构建者的生动案例。综合而言，我们发现这些新能力为我们的研究、工程和产品团队带来了强大的加速效果。



# Cowork: Claude Code for the rest of your work

**Update:** Cowork is now available to Team and Enterprise plans in research preview. (January 23, 2026)

**Update:** Cowork is now available to Pro subscribers in research preview. (January 16, 2026)

## (b) Stack Overflow questions over time

